



Synchronization Issues in Multi-Camera Systems

Problems, root causes, and engineering solutions

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Why Sync Matters

Frame Offset

Cameras capture the same moment at different times — causes **ghosting and tearing** in multi-view footage.

Clock Drift

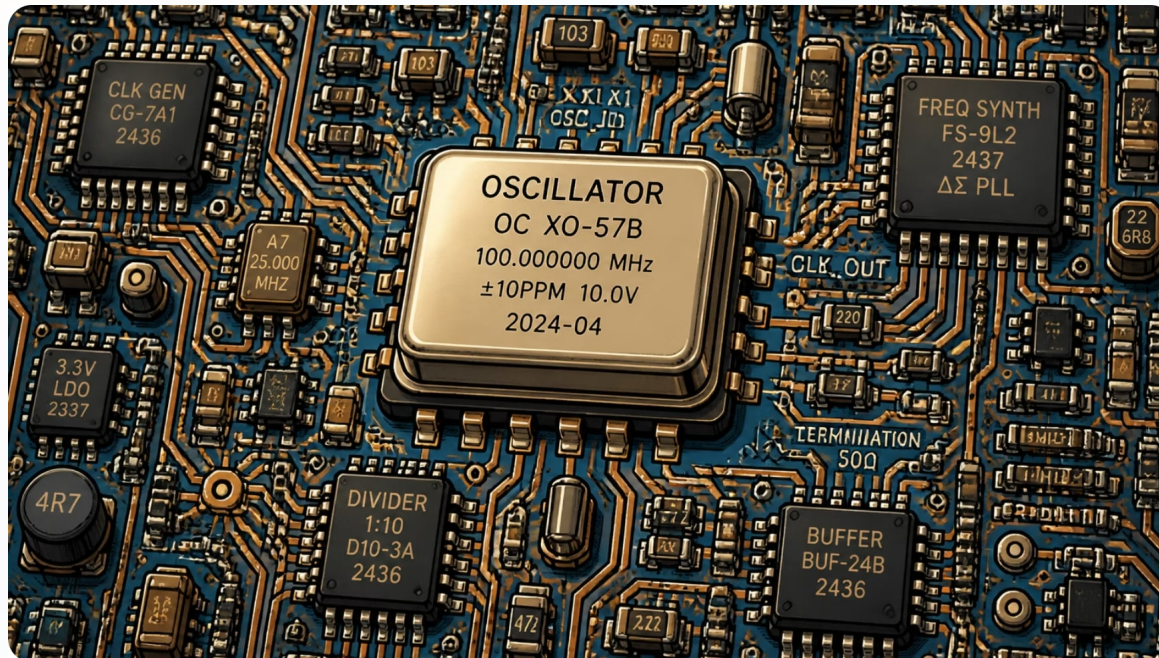
Crystal oscillators drift over time. Even ± 1 ppm accumulates to ~ 86 ms of error per day.

Real Impact

Broken 3D data, **failed post-production alignment**, and glitches during live switching.



Root Causes of Sync Failure



Independent Clocks

Each camera runs its own oscillator — no shared reference means drift is inevitable.

Temperature Changes

Heat shifts oscillator frequency, especially in outdoor or long-running setups.

Mixed Hardware

Different camera brands use incompatible timing interfaces.

Network Jitter

Software sync over standard Ethernet introduces unpredictable delays.

Timecode vs. Genlock

What is Timecode?

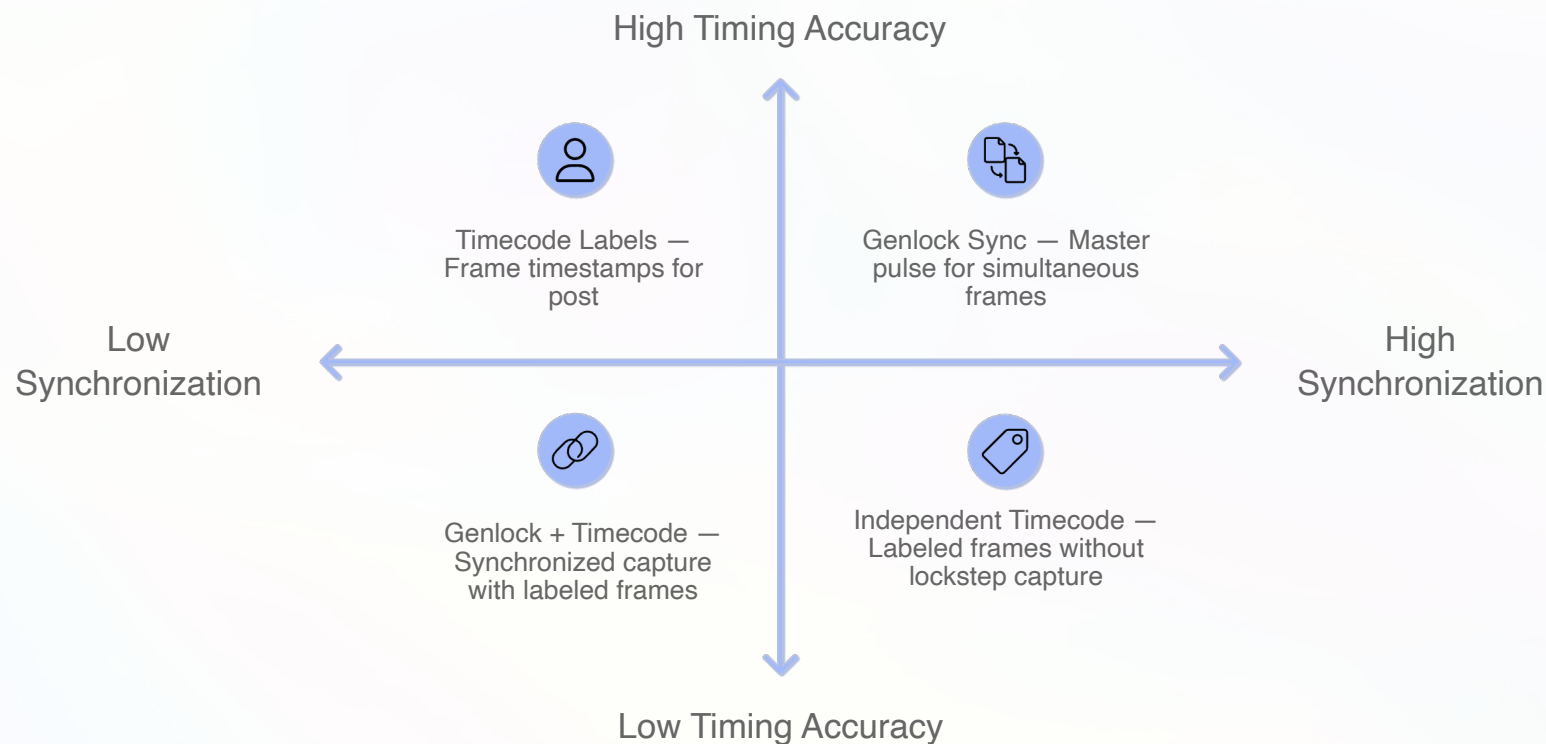
Labels each frame with a timestamp — **HH:MM:SS:FF** — purely metadata. It identifies when a frame occurred but does *not* force cameras to fire together.

- Timecode **names** frames.

What is Genlock?

Sends a **master sync pulse** that forces every camera to begin each new frame at the exact same moment. A hardware-level electrical reference — not metadata.

- Genlock **aligns** frames in real time.



Hardware Sync Solutions



Genlock / Master Sync Generator

Sends **Black Burst or Tri-Level** signals to all cameras via BNC cable — industry standard for broadcast.



FSYNC Trigger Wire

Direct electrical signal shared across cameras. Most accurate method — **sub-millisecond precision**.

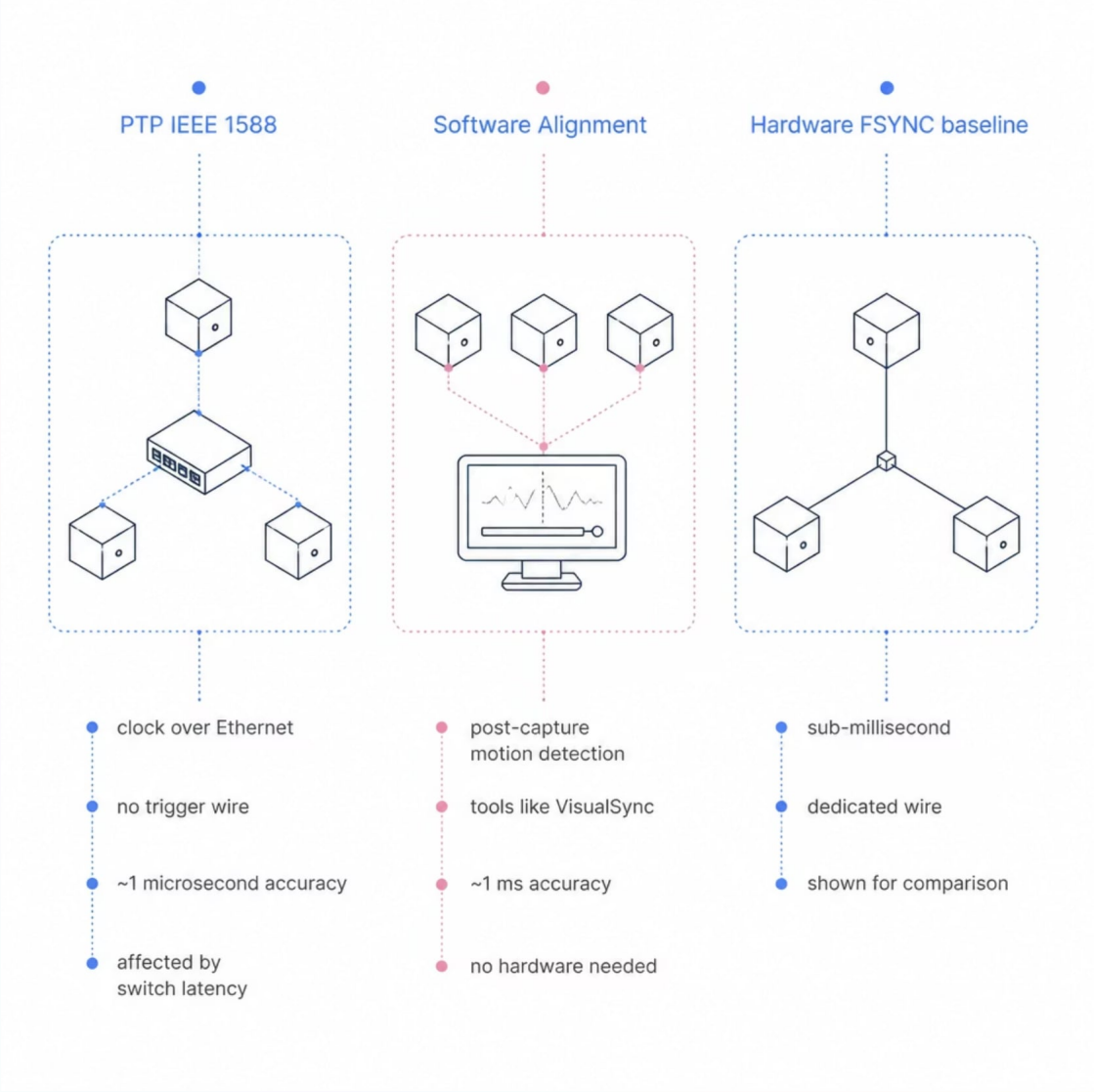


TCXO + VCXO Oscillators

Temperature-compensated clocks with **±0.5 ppm stability** replace standard crystals and reduce drift.



Network-Based Sync Methods



PTP (IEEE 1588)

Shares a common clock over Ethernet — **no trigger wire needed**. Scalable to large distributed setups.

Trade-Off

PTP is less precise than hardware triggers due to **network latency and switch delays**.

Software Alignment

Tools like **VisualSync** detect motion across views to align footage after capture — no special hardware required.

Choosing the Right Sync Method



Broadcast / Live

Genlock + SMPTE timecode

— frame-accurate switching for live production.



Robotics / Machine Vision

Hardware FSYNC — deterministic, sub-millisecond accuracy for real-time systems.



Large Distributed Setups

PTP over Ethernet — scalable with no dedicated sync cabling required.



Post-Production

Software alignment tools — no hardware needed, achieves ~1 ms accuracy.